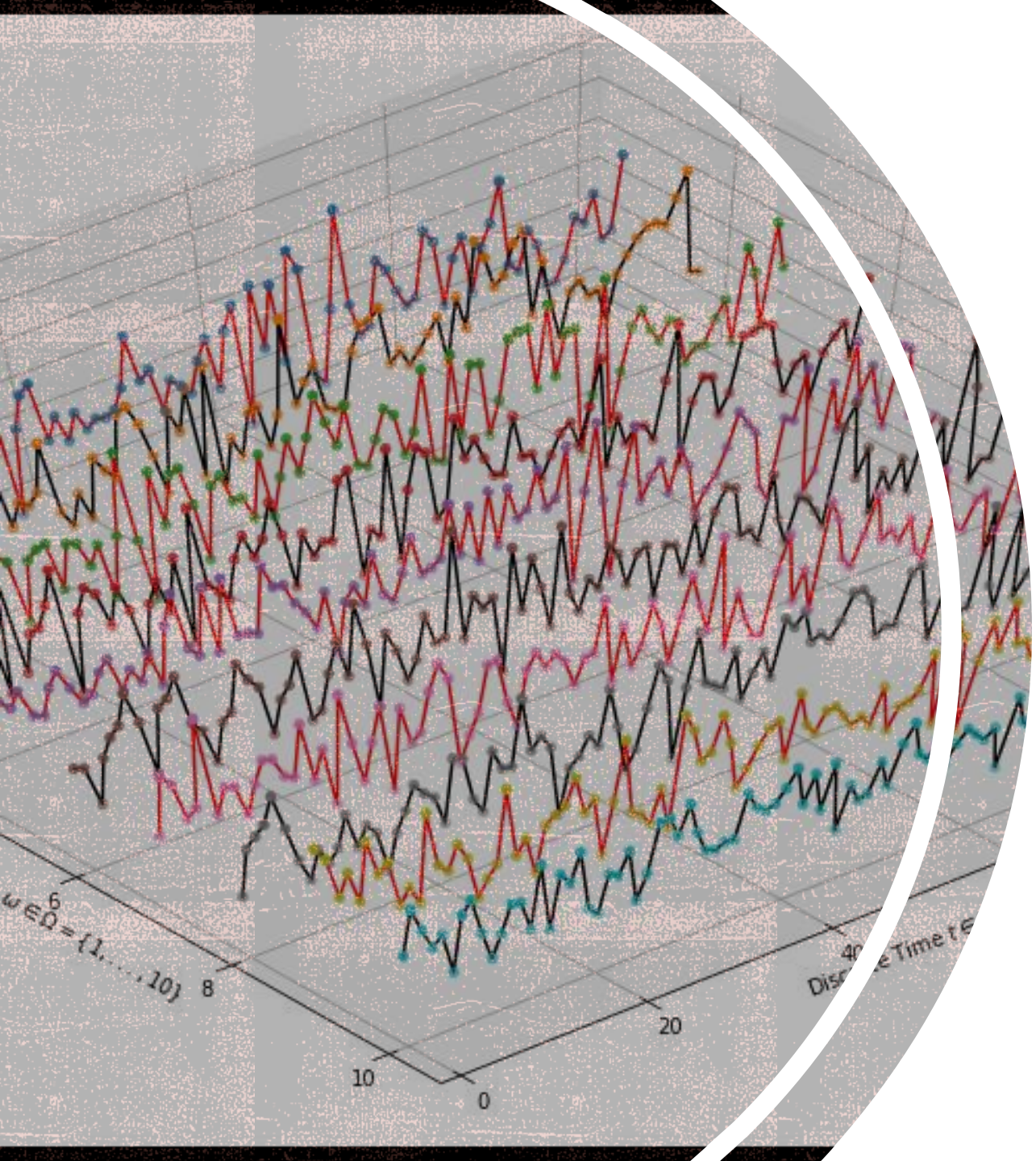


STOCHASTIC PROCESSES IN FINANCE

Valuation for Financial Engineers 6103

Professor David Shimko

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TODAY'S TOOLKIT

- Four types of stochastic processes
  - GBM: Exponential growth random walk
  - ABM: Linear growth random walk
  - MR: Mean-reverting random walk
  - JUMP: Applied to any other random walk
- Time-series statistics
- Manipulation of stochastic processes
- Parameterization
- Optimization

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STOCHASTIC PROCESSES IN FINANCE

Arithmetic random walk

- Continuous time: Arithmetic Brownian Motion
- Used for spreads, interest rates, some currencies
- Can become negative

Geometric random walk

- Continuous time: Geometric Brownian Motion
- Used for asset prices
  - Constant expected log returns and standard deviations
  - Never reaches zero

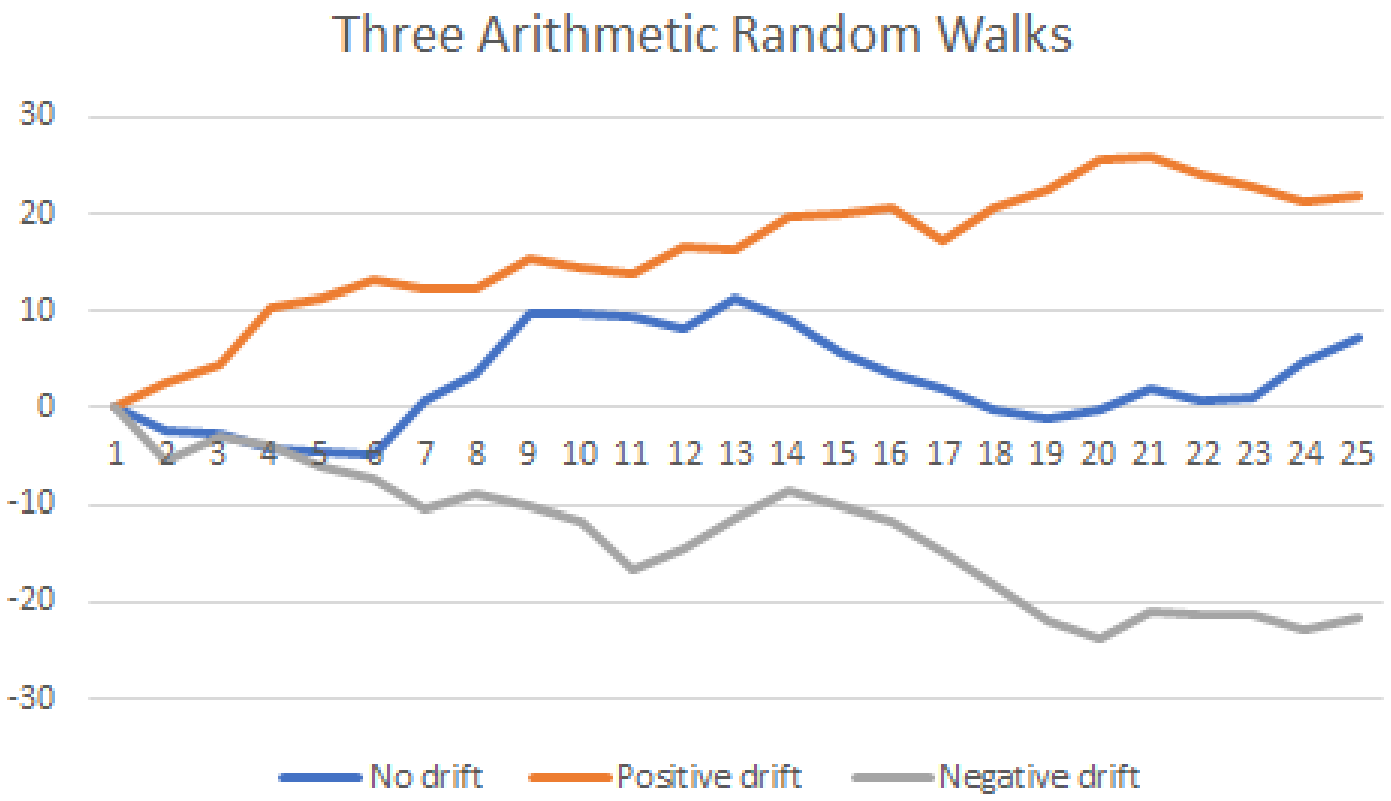
Mean reversion

- Continuous time: Ornstein-Uhlenbeck process
- Used for prices of non-assets
  - Interest rates, Commodities, Foreign exchange
- Negative values occur but are tempered by mean reversion

Jumps

- Used with other models to superimpose discrete, random events

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ARITHMETIC RANDOM WALK DEFINITION

- The stochastic process has a given starting point, e.g. S_0, the stock price at time 0
- The increment in the process is a normal random variable with mean μ and standard deviation σ
- $S_{t+1} = S_t + \mu + \sigma z_1$
- The parameter μ is called the *drift* of the process

DISTRIBUTION OF ARITHMETIC RANDOM WALK OUTCOMES

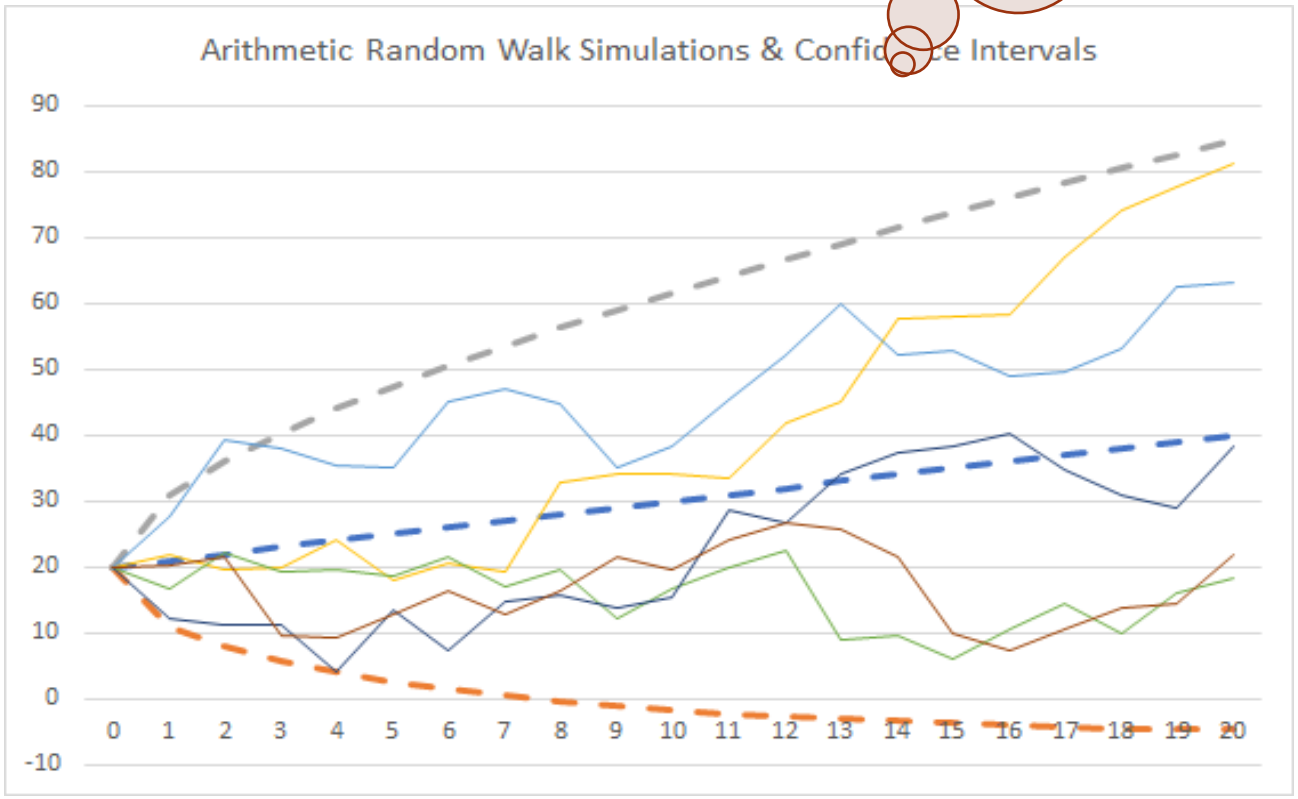
- How can we find the distributions of future values of a process?
- We have $S_{t+1} = S_t + \mu + \sigma z_1$
- What is $S_{t+2} =$ _____?
- Then $S_{t+j} = S_t + j\mu + \sigma \sum_{i=1}^{j} z_{t+i}$
- Since the z valued are iid standard normal, the sum in S_{t+j} is also normally distributed
  - What is the mean of S_{t+j} ? Its variance ?
  - What is the covariance of S_{t+j} with S_{t+k} ?

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ARITHMÉTIC RANDOM WALK

Linearly growing perpetuity?

- $S_{t+1} = S_t + \mu + \sigma z_1$
- $S_{t+j} = S_t + j\mu + \sigma \sum_{i=1}^{j} z_{t+i}$
- $E[S_{t+j}|S_t] = S_t + j\mu$
- $Var[S_{t+j}|S_t] = j\sigma^2$
- $Cov[S_{t+j}, S_{t+m}|S_t] = \sigma^2 \min(j, m)$



Used for spreads; also a foundation for other models

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CLASS EXERCISE

- Generate a 24-month arithmetic random walk for *the natural log of* Amazon stock prices with these parameters in Excel (α and σ are annual). Show at least six simulations on a graph.
  - S=1800, α=20%, σ=25%

- Also show the mean and confidence intervals

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GEOMETRIC RANDOM WALK

- $S_{t+j} = S_t e^{\left(\alpha - \frac{1}{2}\sigma^2\right)j + z_j \sigma \sqrt{j}}$ can be used to generate future values
- $E\left[S_{t+j} \middle| S_t\right] = S_t e^{\alpha j}$
- … exactly as we wanted, with constant expected exponential growth of α.
- Therefore we can express the moments of $ln(S_{t+j})$ to be consistent with this outcome:
- *Log-moments of the Geometric Random Walk*
- $E\left[ln\left(S_{t+j}\right) \middle| S_t\right] = ln(S_t) + \left(\alpha - \frac{\sigma^2}{2}\right)j$
- $Var\left[ln\left(S_{t+j}\right) \middle| S_t\right] = \sigma^2 j$
- $Cov\left[ln\left(S_{t+j}\right), ln(S_{t+m}) \middle| S_t\right] = \sigma^2 \, min(j, m)$

Exponentially growing perpetuity?

Used for asset prices

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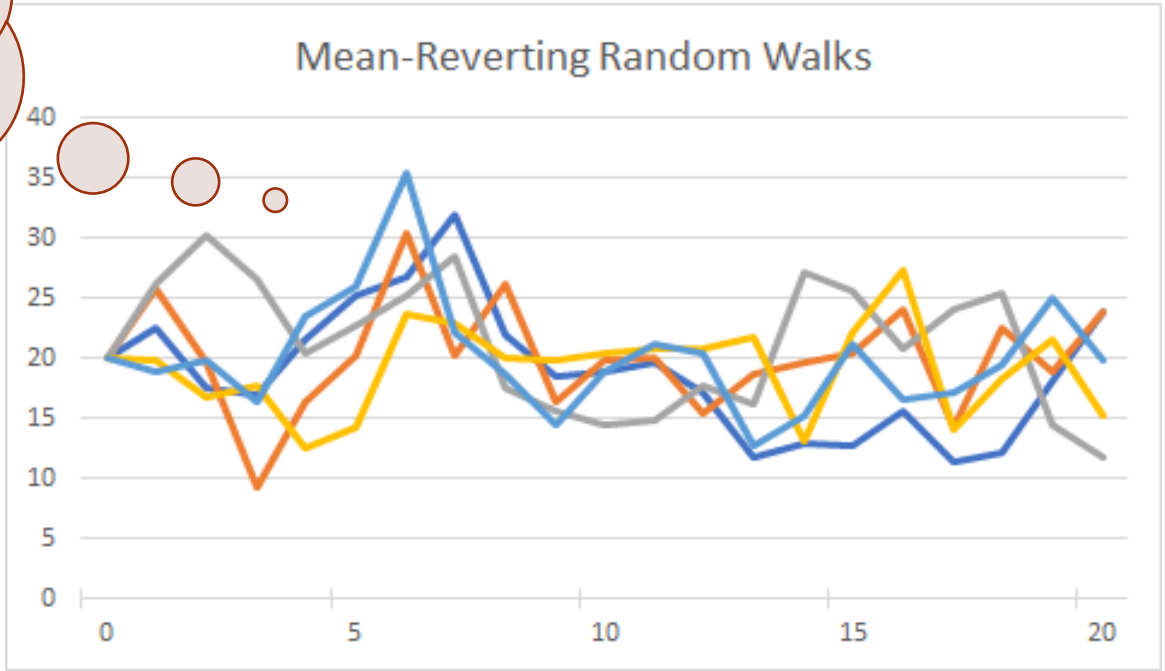
CLASS EXERCISE

- Generate a 24-month geometric random walk for Amazon stock prices with these parameters in Excel (α and σ are annual). Show at least six simulations on a graph.
  - S=1800, α=20%, σ=25%
- Simulate the value of Amazon stock in two years
- Estimate the mean value and verify it is correct

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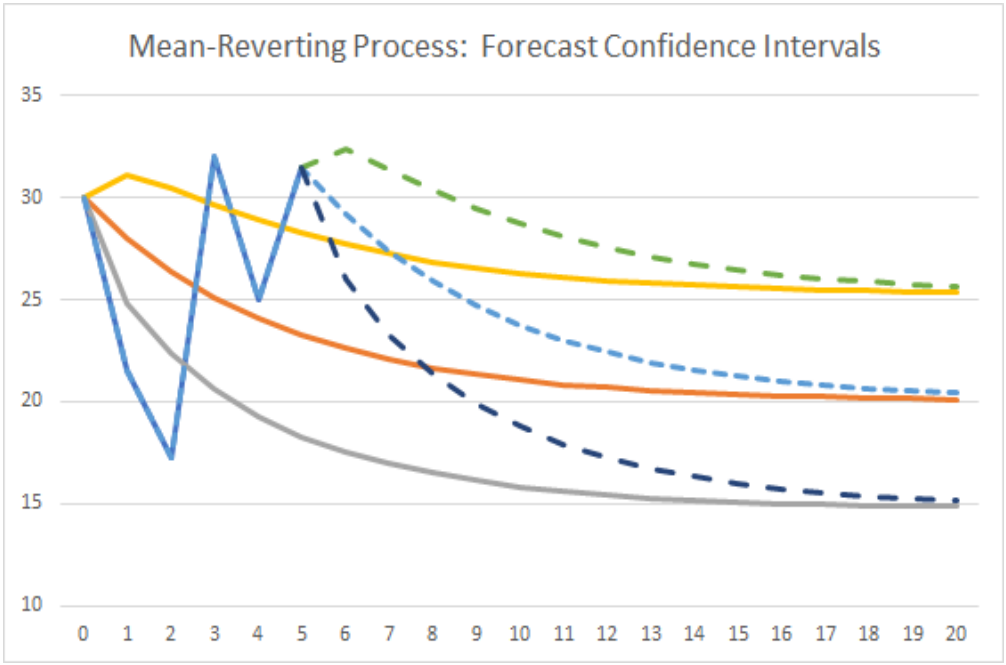
MEAN REVERSION

Mean-reverting perpetuity?



Used for interest rates, commodity prices, some exchange rates

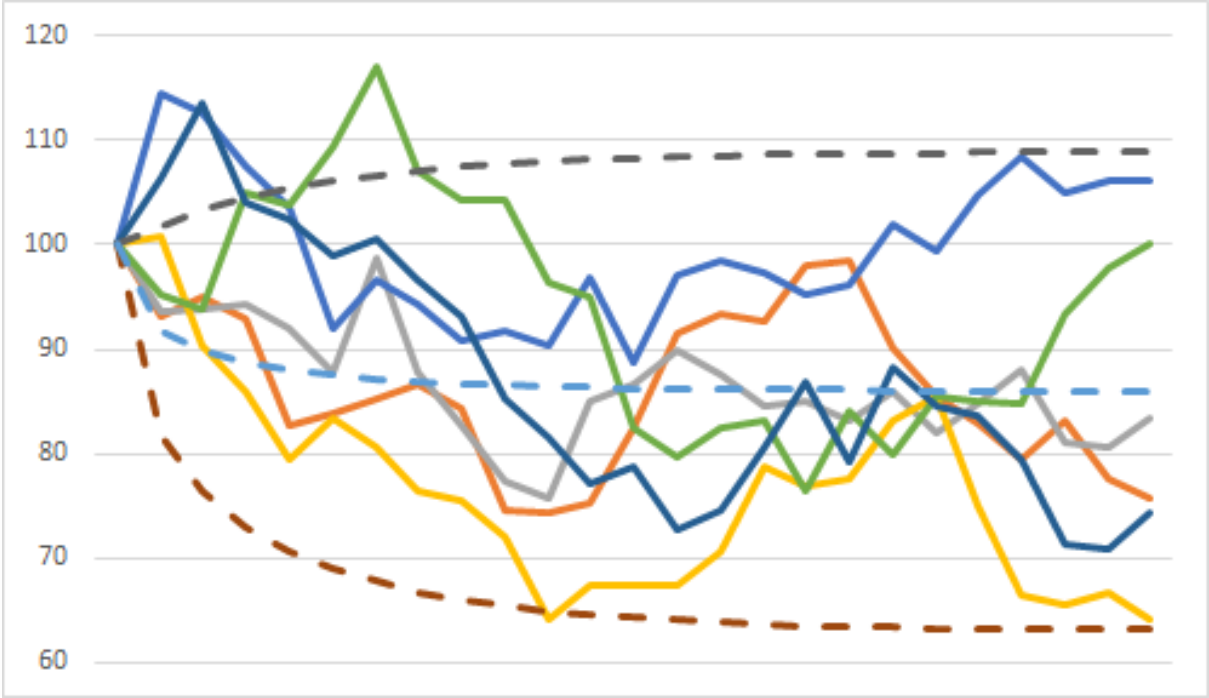
- $S_{t+1} = S_t + \kappa(\mu - S_t) + \sigma z_1$
- $S_{t+j} = \mu + (S_t - \mu)(1-\kappa)^j + \sigma \sum_{i=0}^{j-1} (1-\kappa)^i z_{t+j-i}$
- $E[S_{t+j}|S_t] = \mu + (S_t - \mu)(1-\kappa)^j$
- $Var[S_{t+j}|S_t] = \sigma^2 \left(\frac{1-(1-\kappa)^{2j}}{\kappa(2-\kappa)}\right)$
- $Cov[S_{t+j}, S_{t+m}|S_t] = \sigma^2 \left(\frac{1-(1-\kappa)^{2j}}{\kappa(2-\kappa)}\right)(1-\kappa)^{m-j}\ for\ m \geq j$
- What happens when κ=0?



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EXERCISE

- Simulate a mean-reverting process for 24 months that has initial value 100, long run mean 80, speed of mean reversion 10% per month, volatility per month of $5
- Add the two standard deviation confidence intervals



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JUMP PROCESSES

- Can be added to any process
- At any time. There is a probability π that an amount J will be added to the process
  - J may be a function of the underlying process
  - J may be a random variable itself
  - The value of π may follow a stochastic process itself

- Class exercise
  - Model a mean-reverting process (24 prices) with these parameters and show the graphs
  - Starting value = 100, Long run mean = 80
  - Speed = 20%, Volatility = 5, Jump size = 50, Jump probability = 10%
  - What can you conclude?

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PARAMETER EST. GEOMETRIC

- Calculate log returns from historical data
- Calculate standard deviation of log returns
- **Adjust the drift by adding half the variance**
- Scale as needed
  - Use the T rule for the mean (this is a mean percentage return)
  - Using the $\sqrt{T}$ rule for the volatility (*volatility* is a percentage s.d.)

12 months

Prices	Log of ratio
20	
22	0.09531018
19	-0.14660347
15	-0.23638878
18	0.18232156
19	0.05406722
21	0.10008346
20	-0.04879016
18	-0.10536052
19	0.05406722
22	0.14660347
21	-0.04652002
23	0.09097178

Average	0.011647
Stdev	0.127688

Estimated geometric parameters

Drift	0.237587
Volatility	0.442325

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PARAMETER EST. MR

- Access the MORTGAGE30US file for analysis.
- Test the data for the presence of mean reversion.
- Parameterize the model
  - What is the long run mean estimate?
  - What is the t-statistic for the speed coefficient?
  - Do you believe mean reversion exists?
  - How might you modify the parameterization?
- Extrapolate your model
  - Show the expected future rates based on the model
  - Show one stochastic simulation

- Basic MR equation
- $\Delta X = \kappa(\mu - X)\Delta t + \sigma\varepsilon$
- $E[\Delta X] = \kappa\mu - \kappa X + \sigma\varepsilon$
- Regression
  - Constant = $\kappa\mu$
  - Slope = $-\kappa$
- Run an autoregression
  - Test significance of κ
  - Solve for μ

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COMBINATIONS OF CORRELATED PROCESSES

- Class discussion:
- How would you simulate the value of oil and the value of an oil company jointly?
- How would you determine the correlation to use?

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HISTORICAL SIMULATION

Simulating Daily Stock Returns - Historical simula

	Adj close	Return		Percentile	SORTED
Sep 26, 20	6,643.70	0.59%	0	0.00%	-5.97%
25-Sep-25	6,604.72	-0.50%	1	0.08%	-4.84%
24-Sep-25	6,637.97	-0.28%	2	0.16%	-4.32%
23-Sep-25	6,656.92	-0.55%	3	0.24%	-4.04%
22-Sep-25	6,693.75	0.44%	4	0.32%	-3.88%
19-Sep-25	6,664.36	0.49%	5	0.40%	-3.63%
18-Sep-25	6,631.96	0.48%	6	0.48%	-3.56%
17-Sep-25	6,600.35	-0.10%	7	0.56%	-3.53%
16-Sep-25	6,606.76	-0.13%	8	0.64%	-3.46%
15-Sep-25	6,615.28	0.47%	9	0.72%	-3.37%
12-Sep-25	6,584.29	-0.05%	10	0.80%	-3.25%
11-Sep-25	6,587.47	0.85%	11	0.88%	-3.20%
10-Sep-25	6,532.04	0.30%	12	0.96%	-3.00%
9-Sep-25	6,512.61	0.27%	13	1.04%	-2.95%
8-Sep-25	6,495.15	0.21%	14	1.12%	-2.95%
5-Sep-25	6,481.50	-0.32%	15	1.20%	-2.91%
4-Sep-25	6,502.08	0.83%	16	1.28%	-2.81%
3-Sep-25	6,448.26	0.51%	17	1.36%	-2.80%

Single random return

0.82%

STATISTICS		Annual
Mean	0.0006	15.21%
StDev	0.0108	17.13%
Skew	0.1207	
Kurt	-0.7086	



CONTINUOUS ANALOGUES

- Geometric Brownian motion
  - $dS = \alpha S dt + \sigma S dz$
- Arithmetic Brownian motion
  - $dX = \alpha dt + \sigma dz$
- Ornstein-Uhlenbeck mean-reverting process
  - $dX = \kappa(\mu - X)dt + \sigma dz$
- Wiener process
  - *dz* (dz_t) is a normally distributed infinitesimal, independent for any t pair
  - $E[dz_t] = 0, Var[dz_t] = dt$
  - $dW_t = \alpha dt + \sigma dz$; can be correlated in the usual method for normal distributions
  - $df(W_t, t) = \left(\alpha f_W + \frac{1}{2}\sigma^2 f_{WW} + f_t\right) dt + \sigma f_W dz$ *(Itô's lemma)*

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OPTIMIZATION USING STOCHASTIC PROCESSES

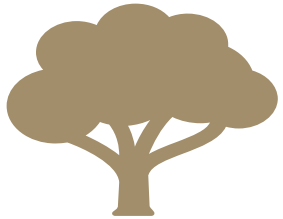
Assume

The price of a tree follows an arithmetic random walk with a given drift and standard deviation

The risk-free interest rate is constant and given

From now until time T in the future, we can "harvest" the tree and collect its value once at any time we choose

Once harvested, the tree is gone



Questions

What does the optimal harvest policy look like?

- (We will assume it is when the tree reaches a given optimized constant value in this example)

What is the optimal harvesting price?

What is the value assuming optimal harvesting?

HOW TO OPTIMIZE?

- Optimization and simulation do not fit well together
- Best to use fixed random numbers (or fixed seeds) to value each asset under different random scenarios
- Decide the form of the optimization rule